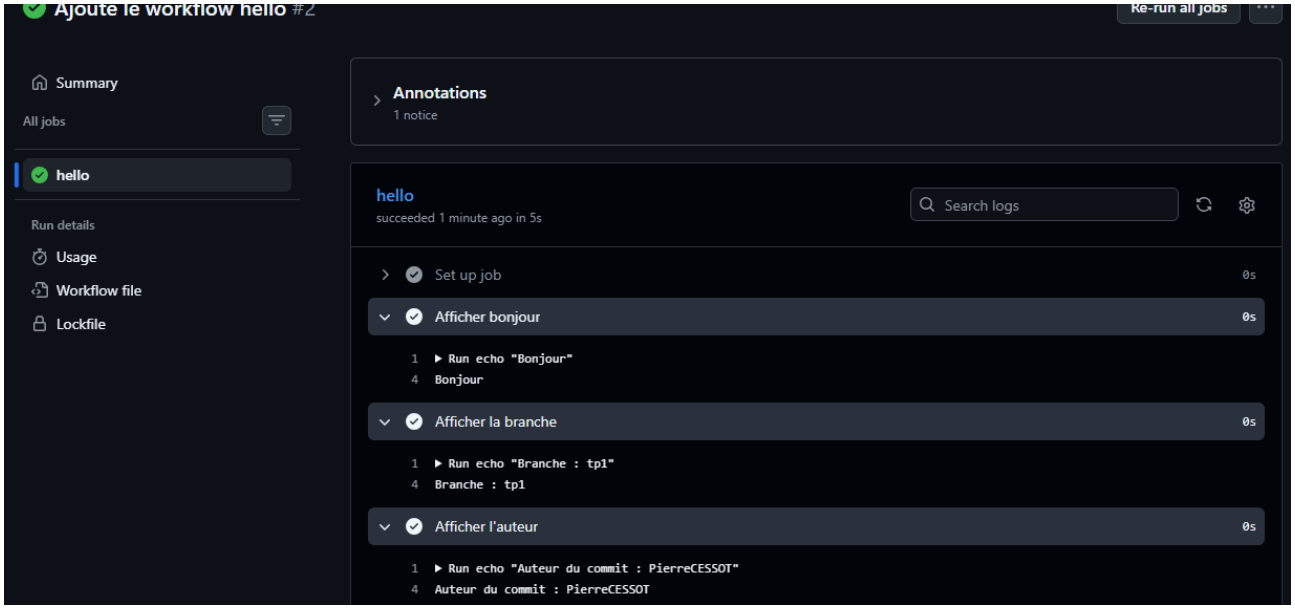
TP1

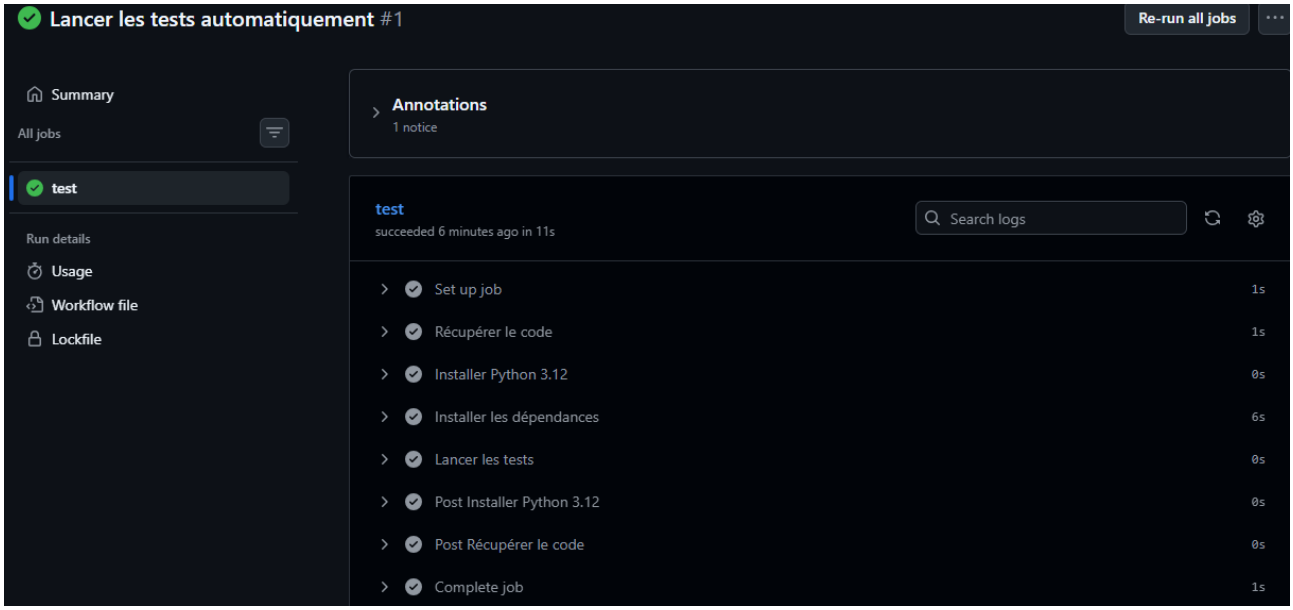



Combien de temps a duré le run « Hello » ?

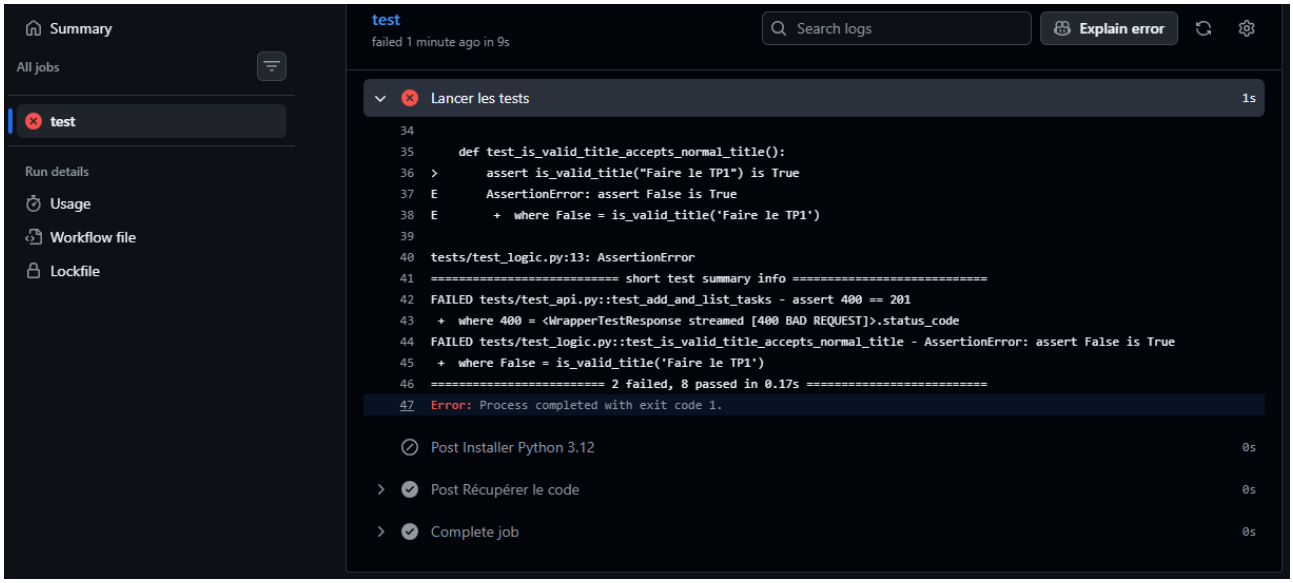
8 secondes pour le run hello

Où voit-on les logs de chaque étape ?

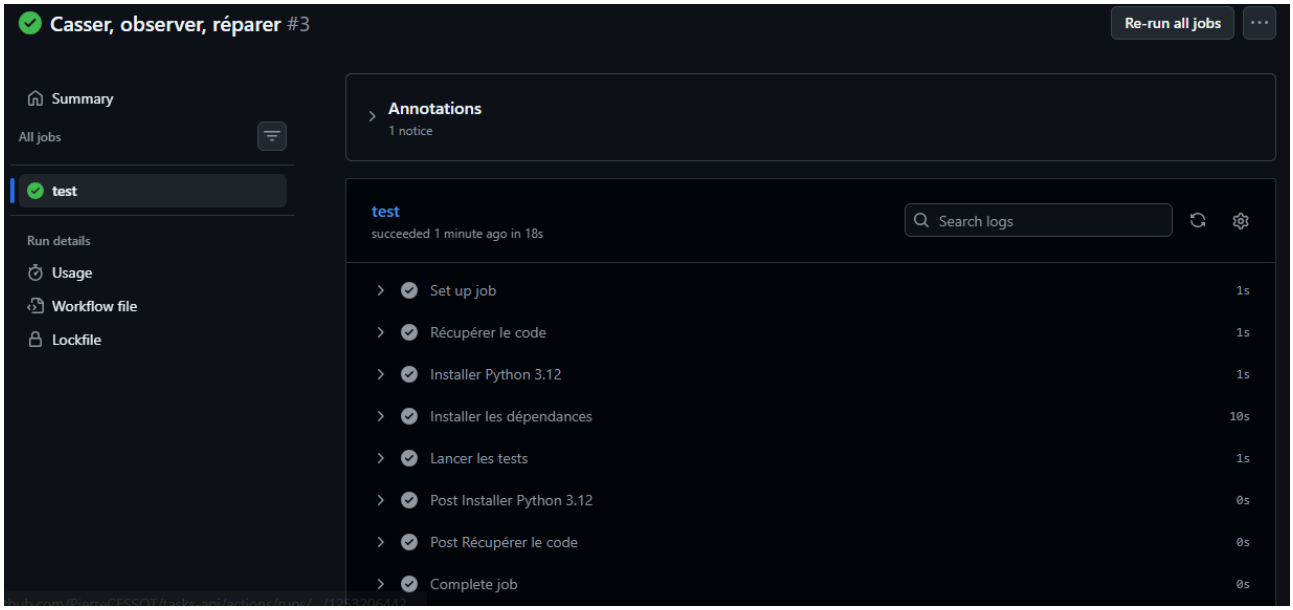
Dans action, le run hello, le job hello



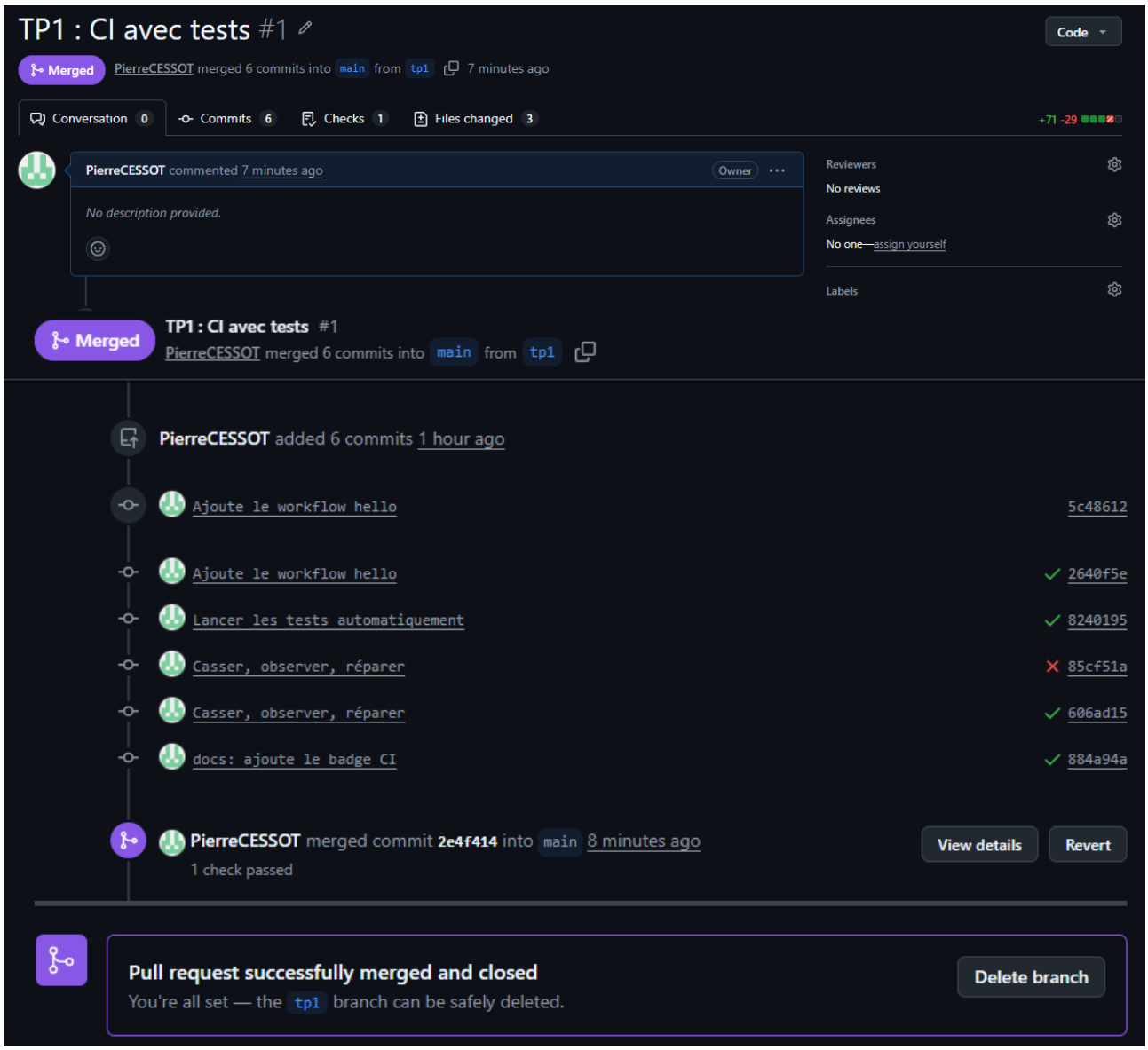
Erreur

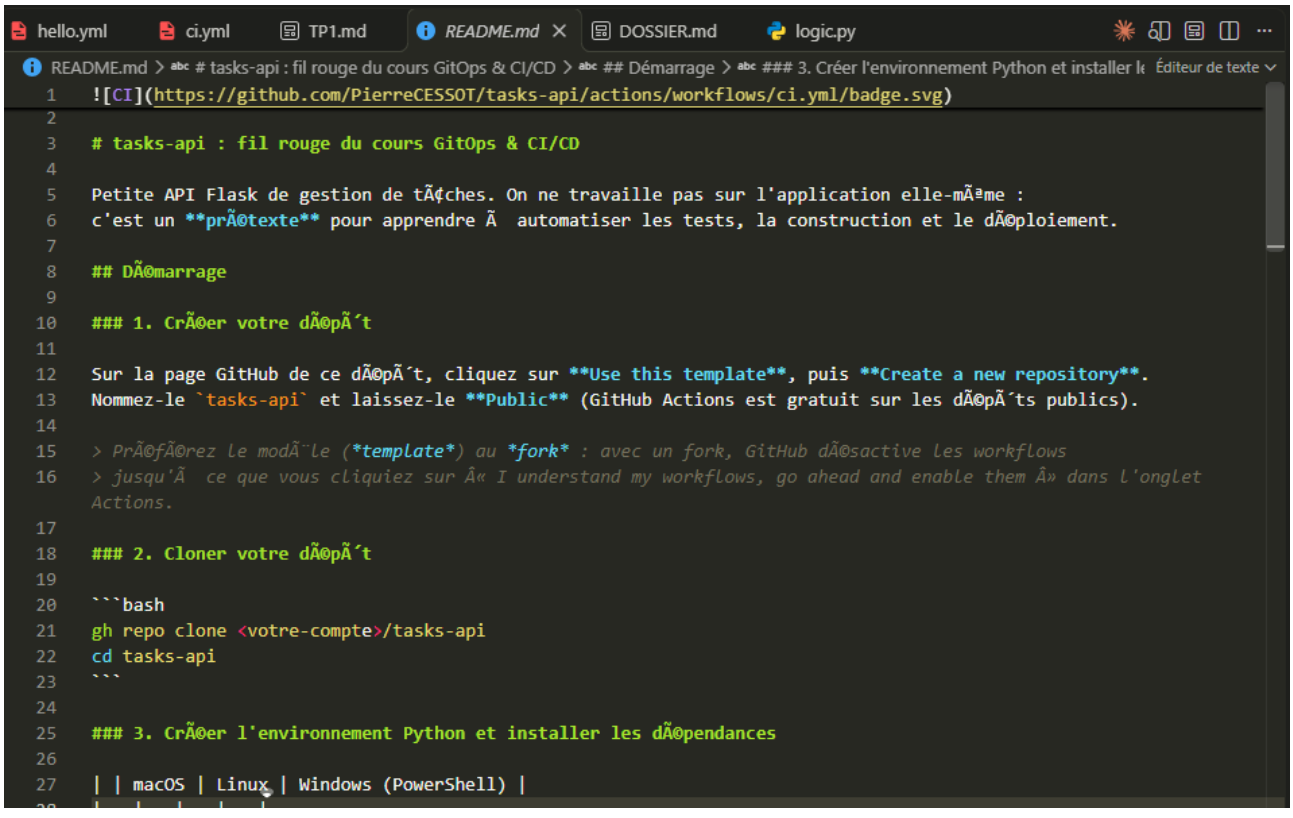


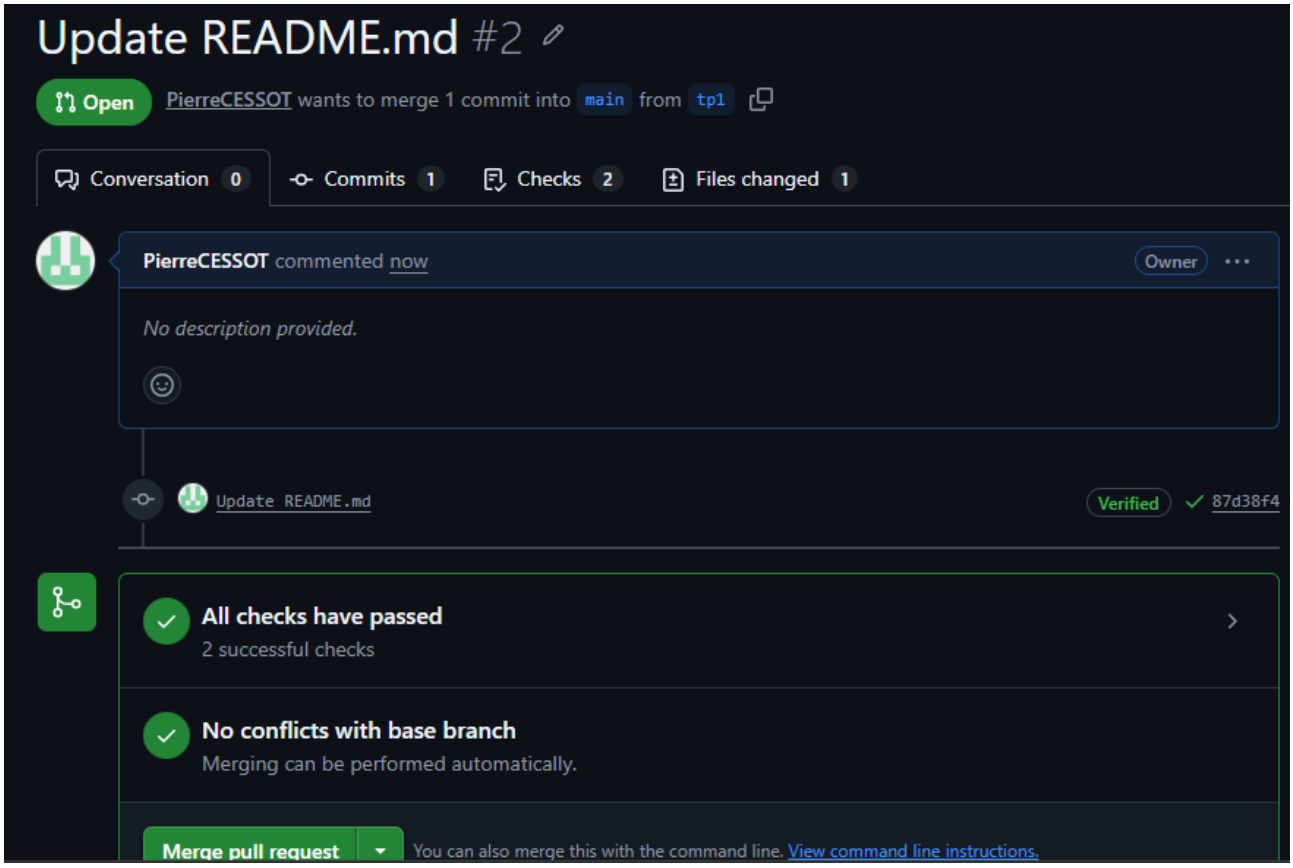

Correction



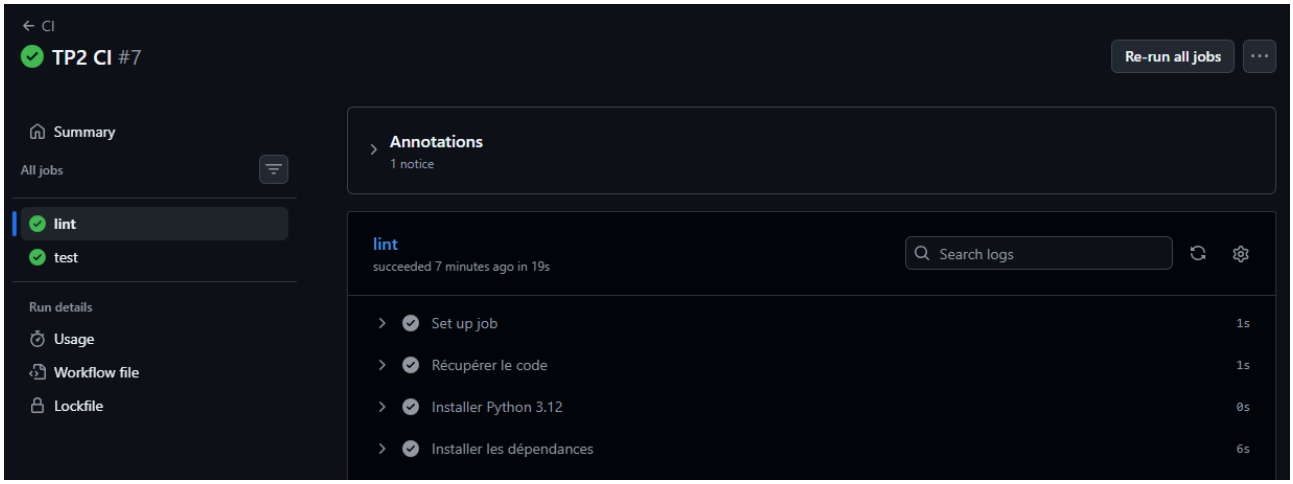

le run pytest a echoué

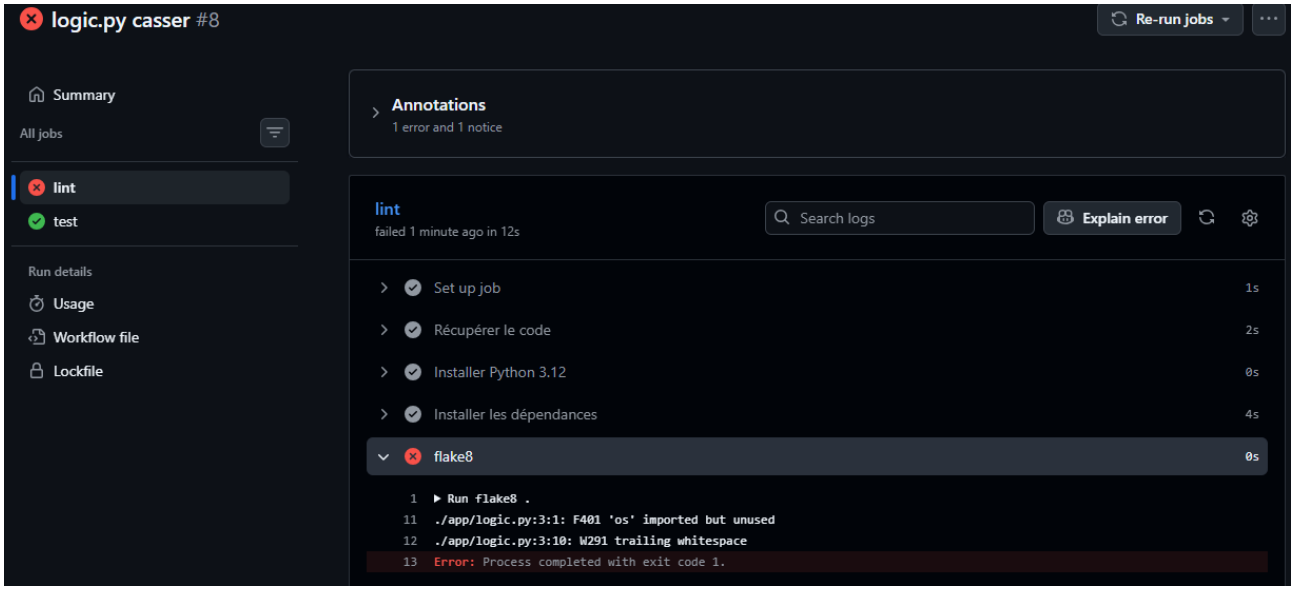

TP1 : CI avec tests #1
Code
Merged
PierreCESSOT merged 6 commits into main from tp1 7 minutes ago
Conversation 0
Commits 6
Checks 1
Files changed 3
+71 -29
PierreCESSOT commented 7 minutes ago
Owner
No description provided.
Reviewers
No reviews
Assignees
No one—assign yourself
Labels
Merged
TP1 : CI avec tests #1
PierreCESSOT merged 6 commits into main from tp1
PierreCESSOT added 6 commits 1 hour ago
Ajoute le workflow hello 5c48612
Ajoute le workflow hello 2640f5e
Lancer les tests automatiquement 8240195
Casser, observer, réparer 85cf51a
Casser, observer, réparer 606ad15
docs: ajoute le badge CI 884a94a
PierreCESSOT merged commit 2e4f414 into main 8 minutes ago
1 check passed
View details
Revert
Pull request successfully merged and closed
You're all set — the tp1 branch can be safely deleted.
Delete branch


hello.yml
ci.yml
TP1.md
README.md
DOSSIER.md
logic.py
README.md > # tasks-api : fil rouge du cours GitOps & CI/CD > ## Démarrage > ### 3. Créer l'environnement Python et installer l
Éditeur de texte
1 ![CI](https://github.com/PierreCESSOT/tasks-api/actions/workflows/ci.yml/badge.svg)
2
3 # tasks-api : fil rouge du cours GitOps & CI/CD
4
5 Petite API Flask de gestion de tÃ¢ches. On ne travaille pas sur l'application elle-mÃªme :
6 c'est un **prÃ©texte** pour apprendre Ã  automatiser les tests, la construction et le dÃ©ploiement.
7
8 ## DÃ©marrage
9
10 ### 1. CrÃ©er votre dÃ©pÃ´t
11
12 Sur la page GitHub de ce dÃ©pÃ´t, cliquez sur **Use this template**, puis **Create a new repository**.
13 Nommez-le `tasks-api` et laissez-le **Public** (GitHub Actions est gratuit sur les dÃ©pÃ´ts publics).
14
15 > *PrÃ©fÃ©rez le modÃ¨le (*template*) au *fork* : avec un fork, GitHub dÃ©sactive les workflows*
16 > *jusqu'Ã  ce que vous cliquiez sur Â« I understand my workflows, go ahead and enable them Â» dans l'onglet Actions.*
17
18 ### 2. Cloner votre dÃ©pÃ´t
19
20 ```bash
21 gh repo clone <votre-compte>/tasks-api
22 cd tasks-api
23 ```
24
25 ### 3. CrÃ©er l'environnement Python et installer les dÃ©pendances
26
27 | | macOS | Linux | Windows (PowerShell) |

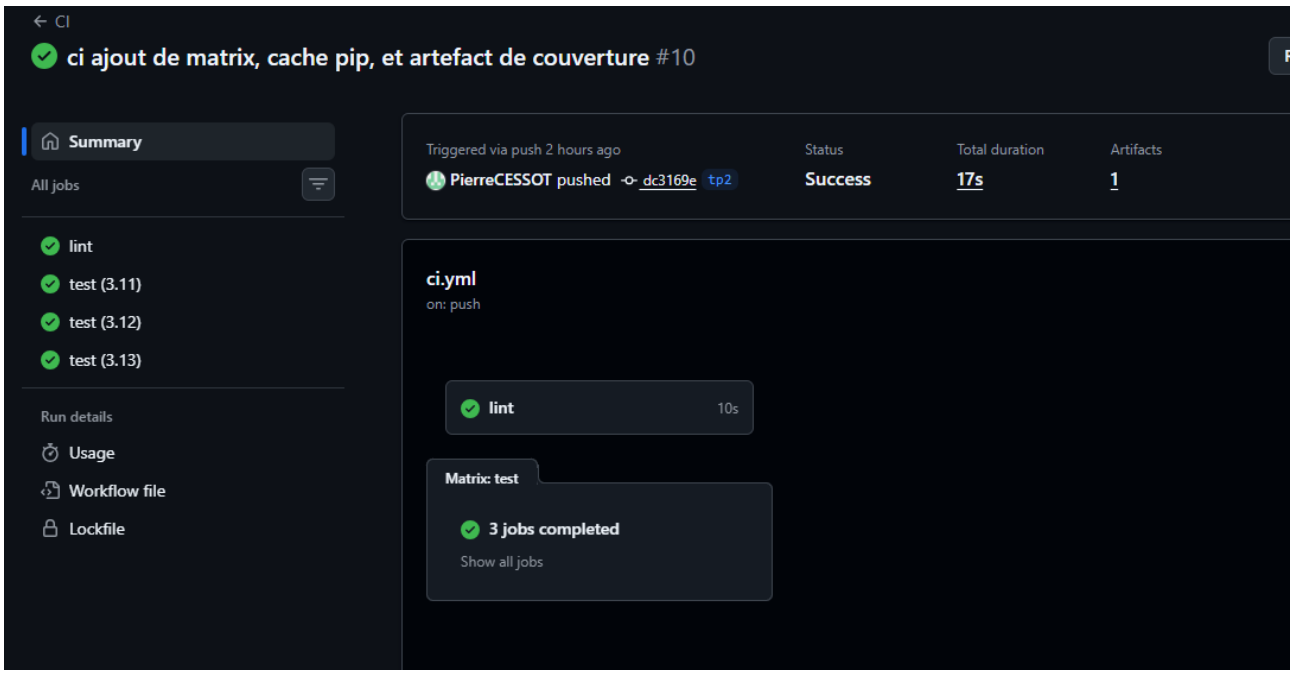

Update README.md #2
Open
PierreCESSOT wants to merge 1 commit into main from tp1
Conversation 0
Commits 1
Checks 2
Files changed 1
PierreCESSOT commented now
Owner
No description provided.
Update README.md
Verified
87d38f4
All checks have passed
2 successful checks
No conflicts with base branch
Merging can be performed automatically.
Merge pull request
You can also merge this with the command line. View command line instructions.

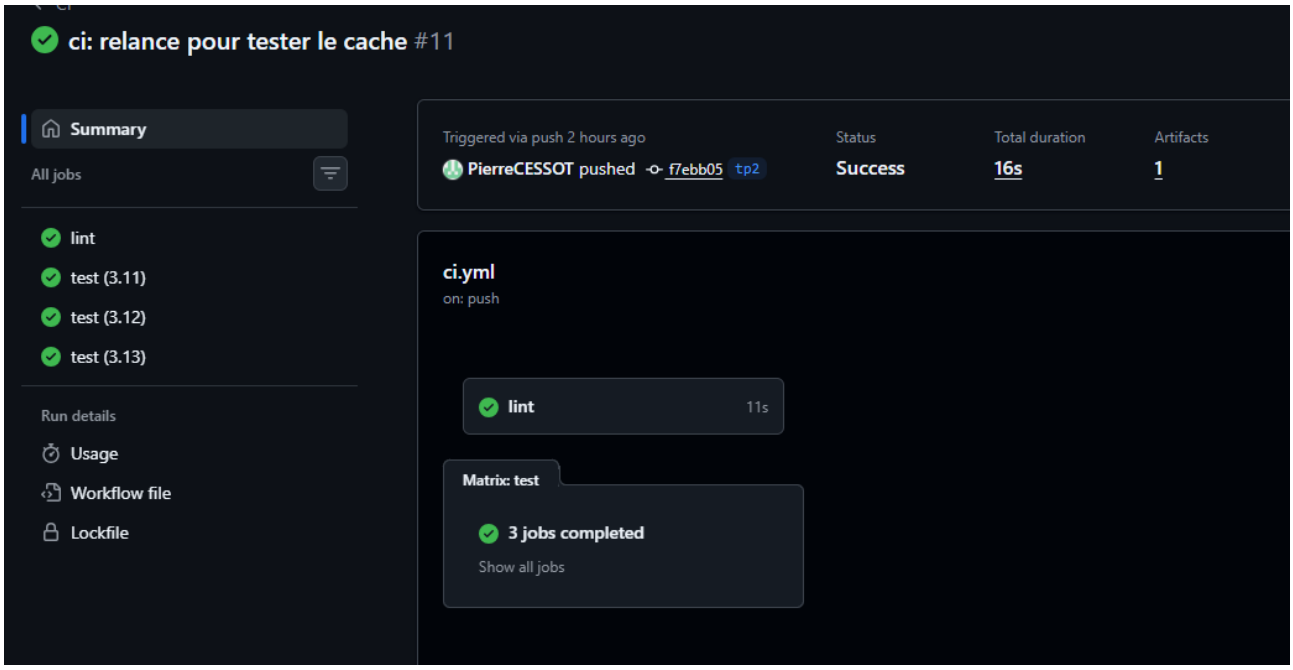
TP2

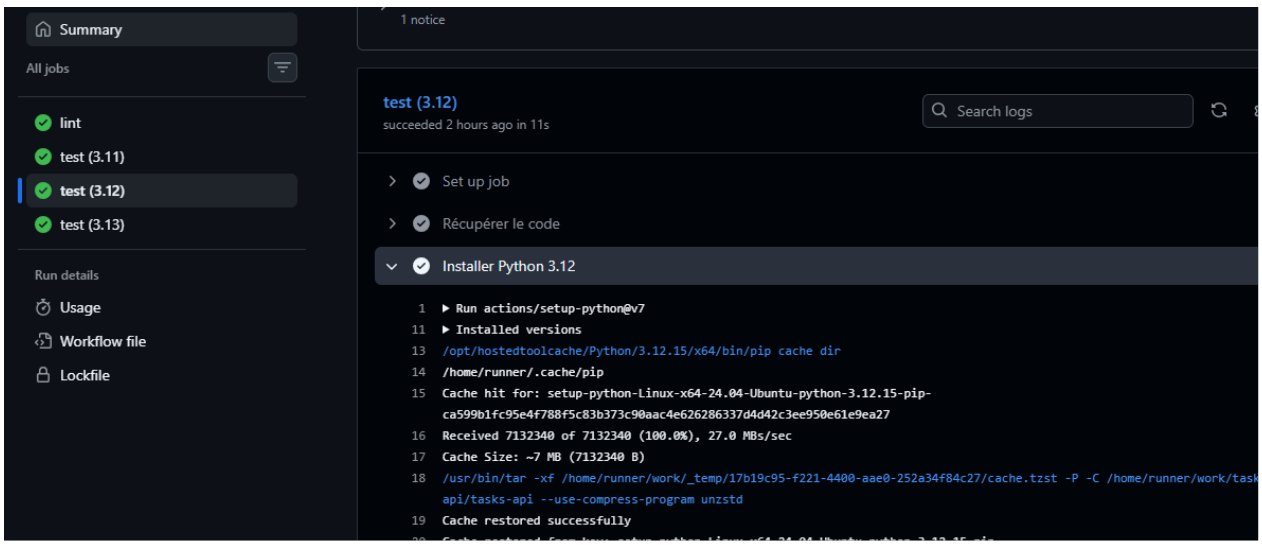

CI
TP2 CI #7
Re-run all jobs
Summary
All jobs
lint
test
Run details
Usage
Workflow file
Lockfile
Annotations
1 notice
lint
succeeded 7 minutes ago in 19s
Search logs
Set up job 1s
Récupérer le code 1s
Installer Python 3.12 0s
Installer les dépendances 6s

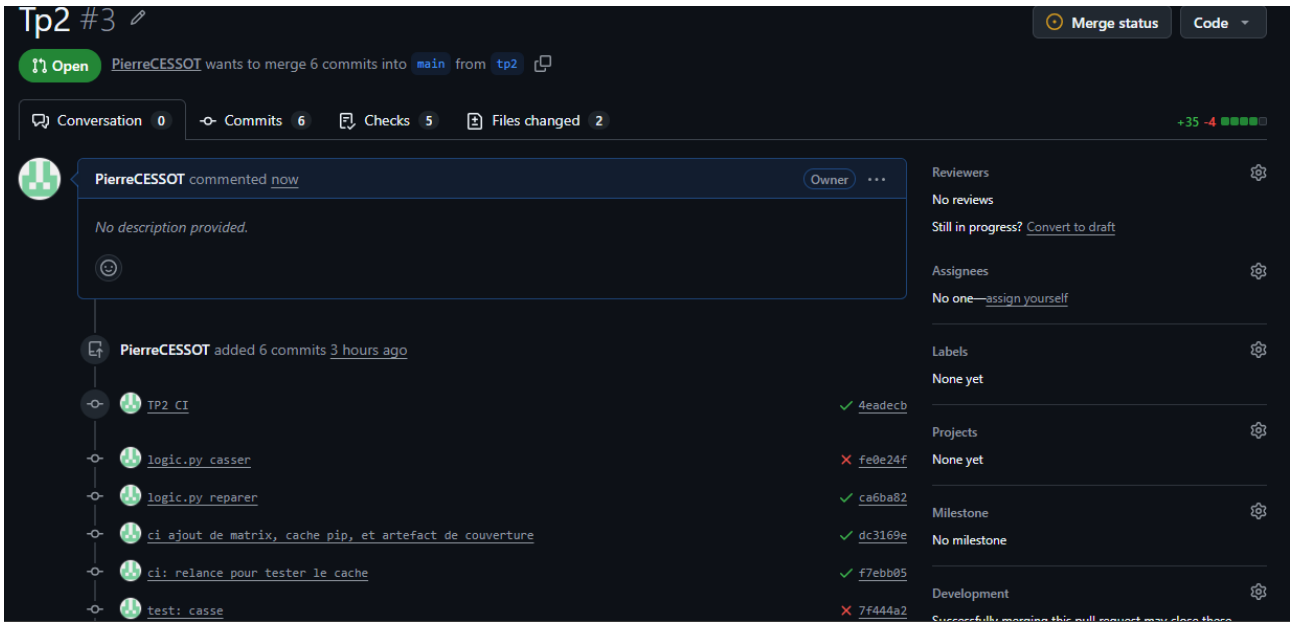

logic.py casser #8
Re-run jobs
Summary
All jobs
lint
test
Run details
Usage
Workflow file
Lockfile
Annotations
1 error and 1 notice
lint
failed 1 minute ago in 12s
Search logs
Explain error
Set up job 1s
Récupérer le code 2s
Installer Python 3.12 0s
Installer les dépendances 4s
flake8 0s
1 ▶ Run flake8 .
11 ./app/logic.py:3:1: F401 'os' imported but unused
12 ./app/logic.py:3:10: W291 trailing whitespace
13 Error: Process completed with exit code 1.

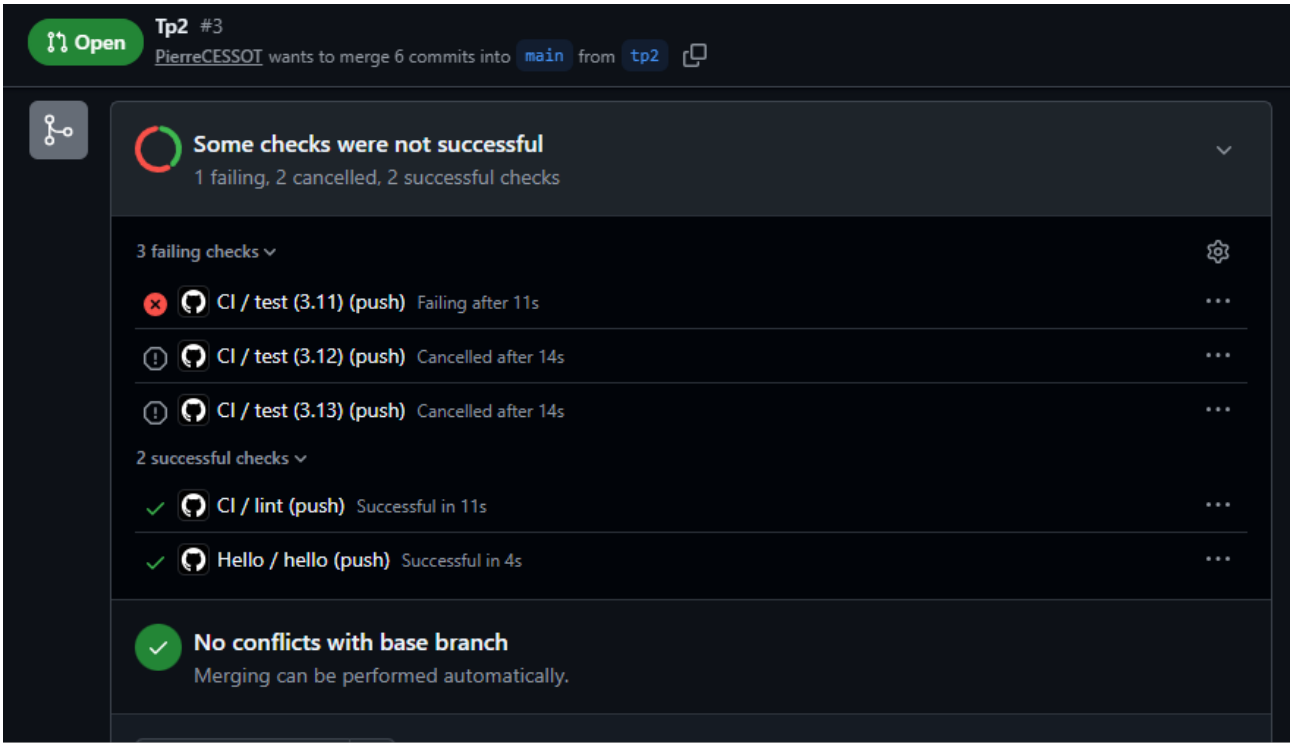

CI
logic.py reparer #9
Re-run all jobs
Summary
All jobs
lint
test
Run details
Usage
Workflow file
Lockfile
Annotations
1 notice
lint
succeeded now in 21s
Search logs
Set up job 0s
Récupérer le code 0s
Installer Python 3.12 1s
Installer les dépendances 5s

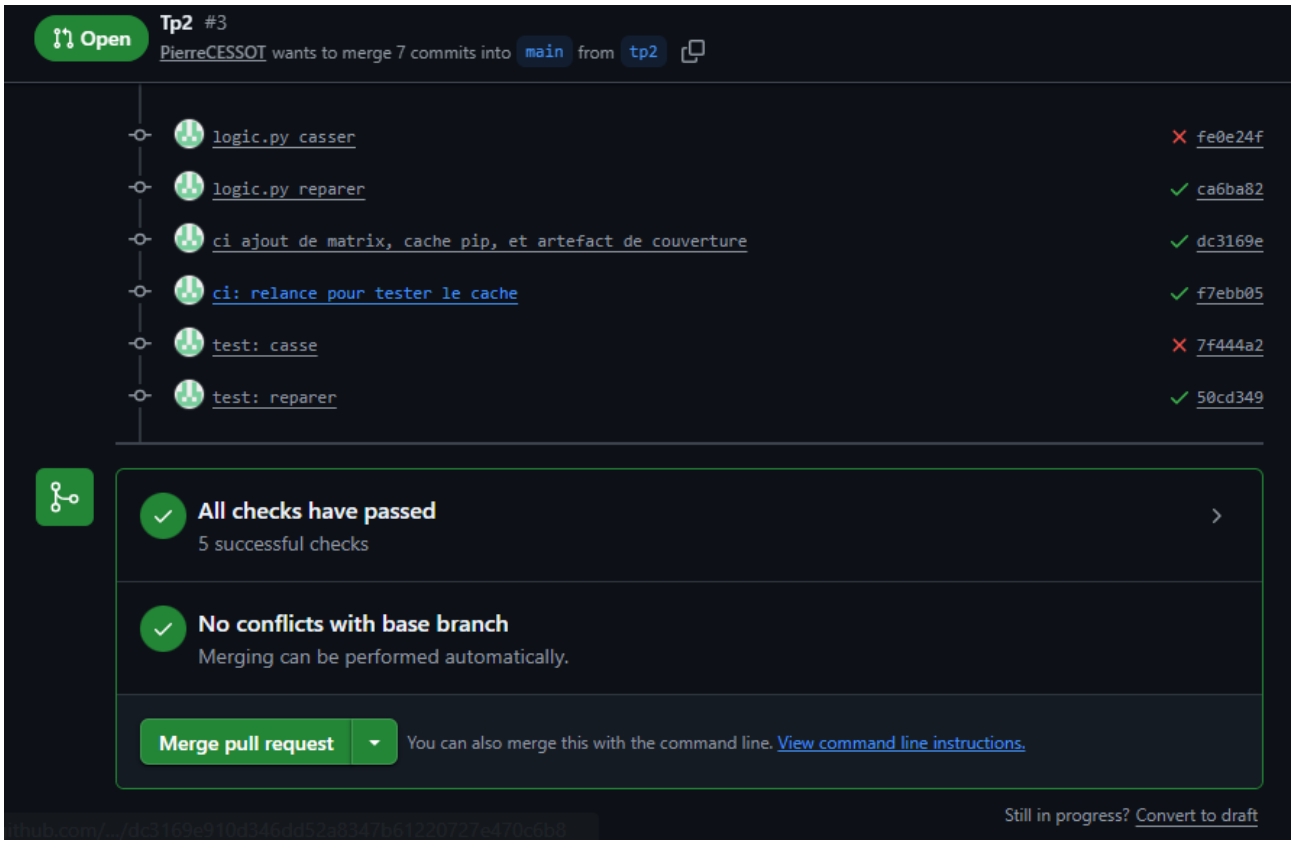

CI
ci ajout de matrix, cache pip, et artefact de couverture #10
Summary
All jobs
lint
test (3.11)
test (3.12)
test (3.13)
Run details
Usage
Workflow file
Lockfile
Triggered via push 2 hours ago
PierreCESSOT pushed dc3169e tp2
Status
Success
Total duration
17s
Artifacts
1
ci.yml
on: push
lint 10s
Matrix: test
3 jobs completed
Show all jobs

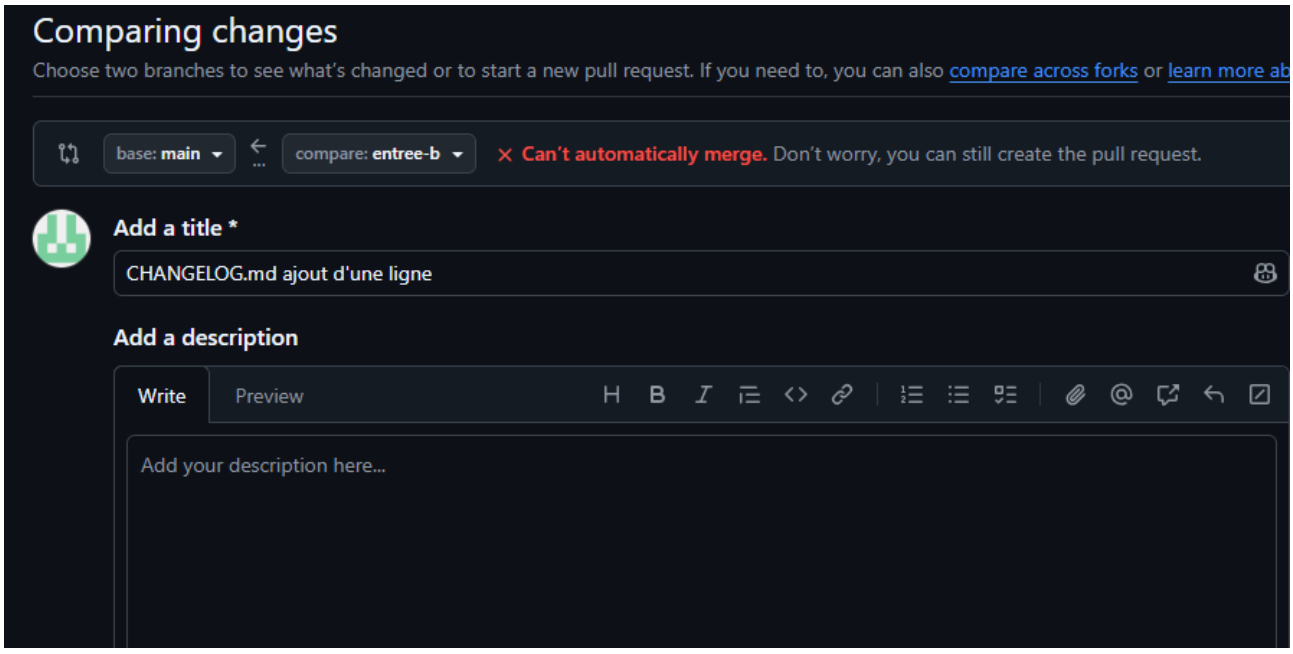

ci: relance pour tester le cache #11
Summary
All jobs
lint
test (3.11)
test (3.12)
test (3.13)
Run details
Usage
Workflow file
Lockfile
Triggered via push 2 hours ago
PierreCESSOT pushed f7ebb05 tp2
Status
Success
Total duration
16s
Artifacts
1
ci.yml
on: push
lint 11s
Matrix: test
3 jobs completed
Show all jobs

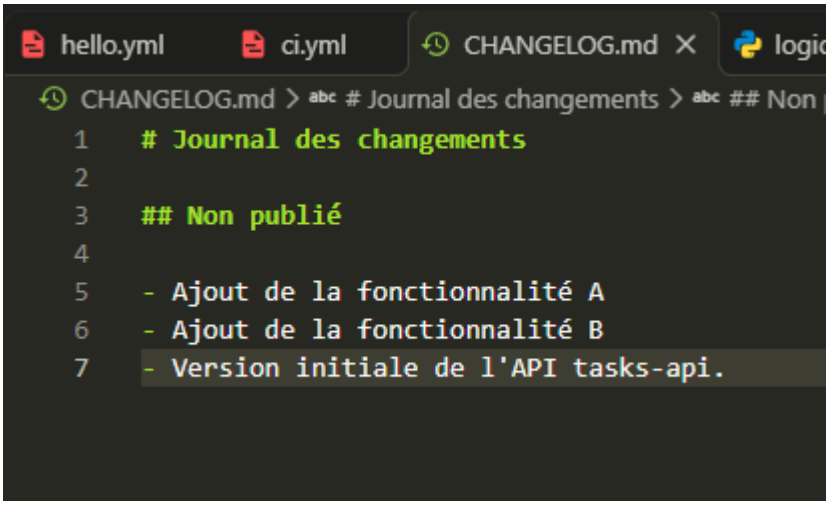

Summary
All jobs
lint
test (3.11)
test (3.12)
test (3.13)
Run details
Usage
Workflow file
Lockfile
1 notice
test (3.12)
succeeded 2 hours ago in 11s
Search logs
Set up job
Récupérer le code
Installer Python 3.12
1 ▶ Run actions/setup-python@v7
11 ▶ Installed versions
13 /opt/hostedtoolcache/Python/3.12.15/x64/bin/pip cache dir
14 /home/runner/.cache/pip
15 Cache hit for: setup-python-Linux-x64-24.04-Ubuntu-python-3.12.15-pip-ca599b1fc95e4f788f5c83b373c90aac4e626286337d4d42c3ee950e61e9ea27
16 Received 7132340 of 7132340 (100.0%), 27.0 MBs/sec
17 Cache Size: ~7 MB (7132340 B)
18 /usr/bin/tar -xf /home/runner/work/_temp/17b19c95-f221-4400-aae0-252a34f84c27/cache.tzst -P -C /home/runner/work/task api/tasks-api --use-compress-program unzstd
19 Cache restored successfully

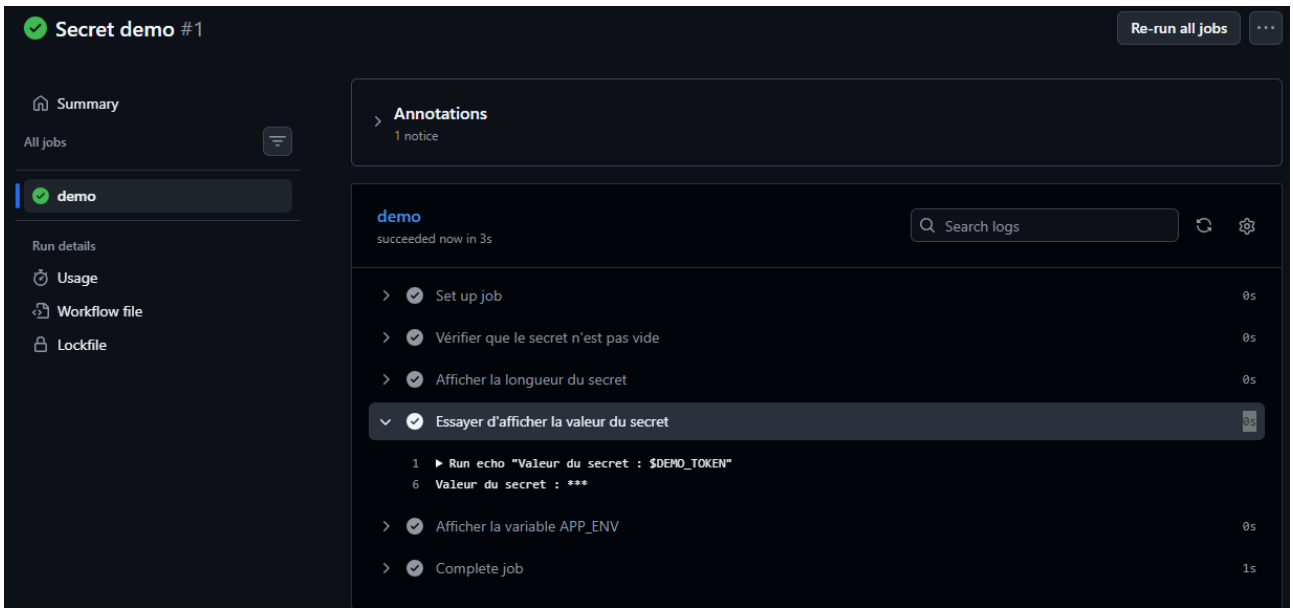

Tp2 #3
Merge status
Code
Open
PierreCESSOT wants to merge 6 commits into main from tp2
Conversation 0
Commits 6
Checks 5
Files changed 2
+35 -4
PierreCESSOT commented now
Owner
No description provided.
PierreCESSOT added 6 commits 3 hours ago
TP2 CI
4eadecb
logic.py casser
fe0e24f
logic.py reparer
ca6ba82
ci ajout de matrix, cache pip, et artefact de couverture
dc3169e
ci: relance pour tester le cache
f7ebb05
test: casse
7f444a2
Reviewers
No reviews
Still in progress? Convert to draft
Assignees
No one—assign yourself
Labels
None yet
Projects
None yet
Milestone
No milestone
Development

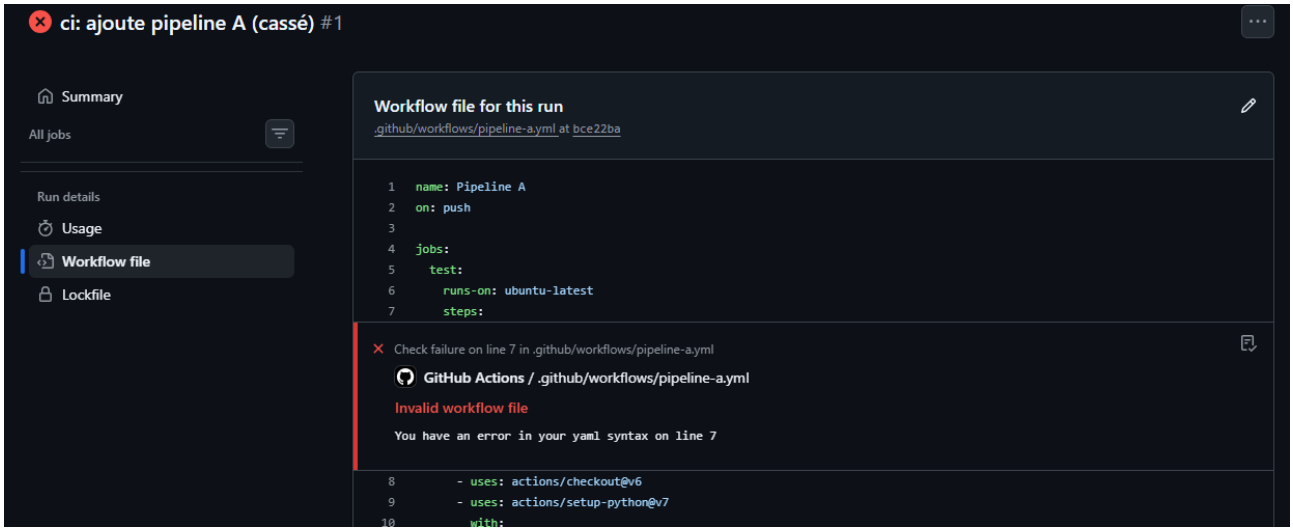

Open
Tp2 #3
PierreCESSOT wants to merge 6 commits into main from tp2
Some checks were not successful
1 failing, 2 cancelled, 2 successful checks
3 failing checks
CI / test (3.11) (push) Failing after 11s
CI / test (3.12) (push) Cancelled after 14s
CI / test (3.13) (push) Cancelled after 14s
2 successful checks
CI / lint (push) Successful in 11s
Hello / hello (push) Successful in 4s
No conflicts with base branch
Merging can be performed automatically.

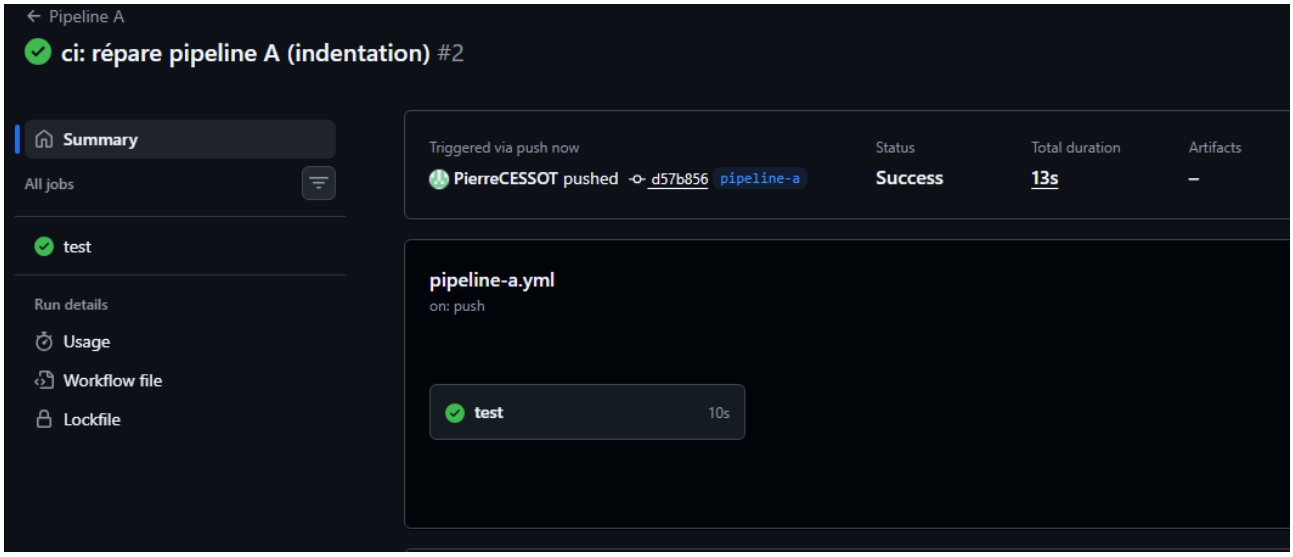

Open
Tp2 #3
PierreCESSOT wants to merge 7 commits into main from tp2
logic.py casser
fe0e24f
logic.py reparer
ca6ba82
ci ajout de matrix, cache pip, et artefact de couverture
dc3169e
ci: relance pour tester le cache
f7ebb05
test: casse
7f444a2
test: reparer
50cd349
All checks have passed
5 successful checks
No conflicts with base branch
Merging can be performed automatically.
Merge pull request
You can also merge this with the command line. View command line instructions.
Still in progress? Convert to draft

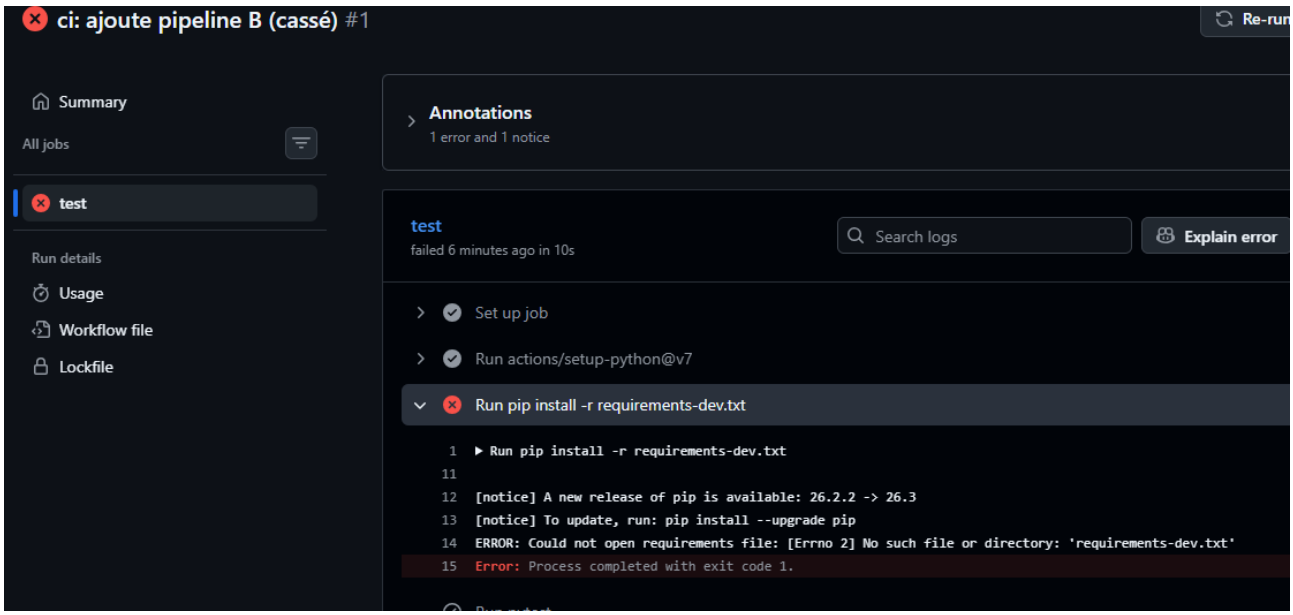

Comparing changes
Choose two branches to see what's changed or to start a new pull request. If you need to, you can also compare across forks or learn more ab
base: main
compare: entree-b
Can't automatically merge. Don't worry, you can still create the pull request.
Add a title *
CHANGELOG.md ajout d'une ligne
Add a description
Write
Preview
Add your description here...

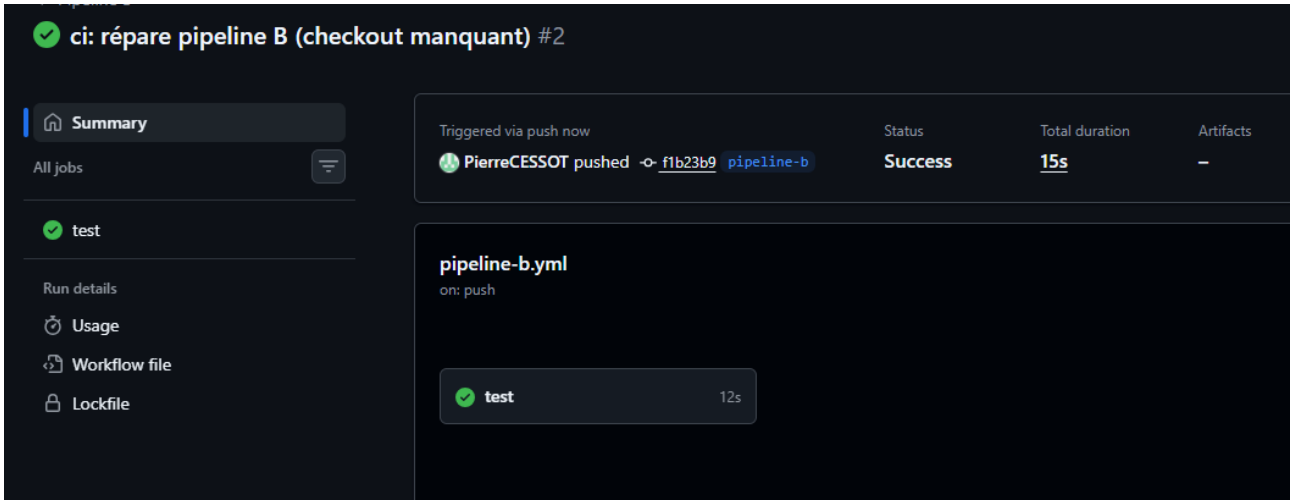

hello.yml
ci.yml
CHANGELOG.md
CHANGELOG.md > # Journal des changements > ## Non
1 # Journal des changements
2
3 ## Non publié
4
5 - Ajout de la fonctionnalité A
6 - Ajout de la fonctionnalité B
7 - Version initiale de l'API tasks-api.

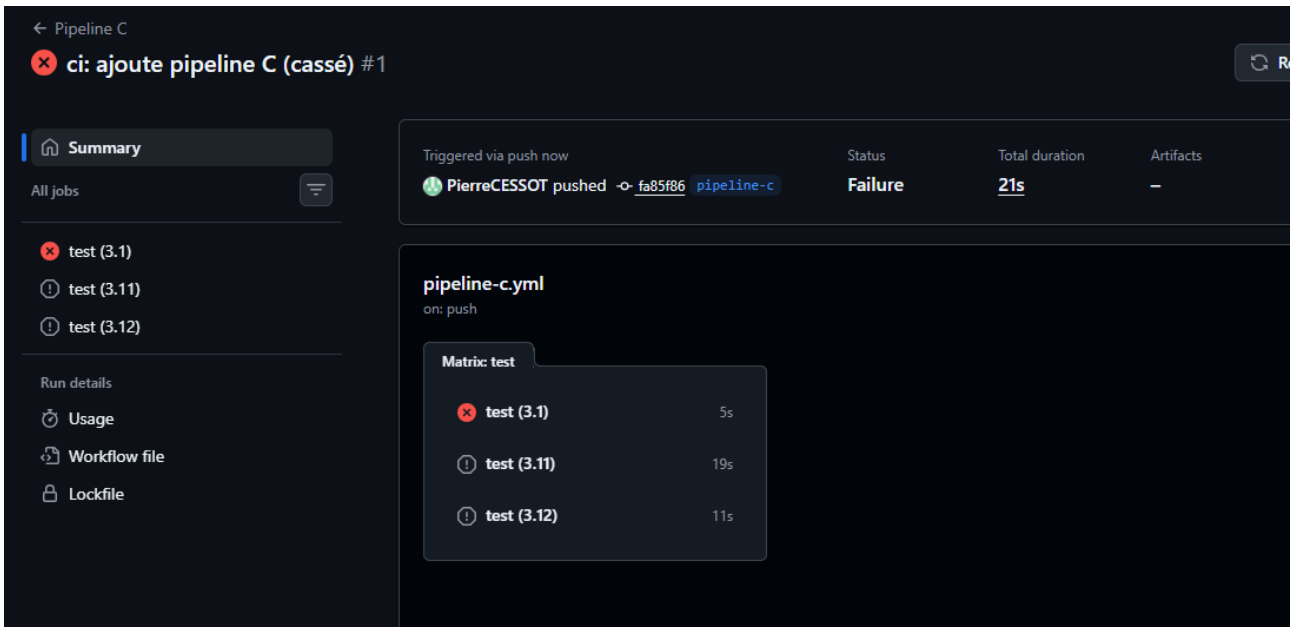

Secret demo #1
Re-run all jobs
Summary
All jobs
demo
Run details
Usage
Workflow file
Lockfile
Annotations
1 notice
demo
succeeded now in 3s
Search logs
Set up job 0s
Vérifier que le secret n'est pas vide 0s
Afficher la longueur du secret 0s
Essayer d'afficher la valeur du secret
1 ▶ Run echo "Valeur du secret : $DEMO_TOKEN"
6 Valeur du secret : ***
Afficher la variable APP_ENV 0s
Complete job 1s

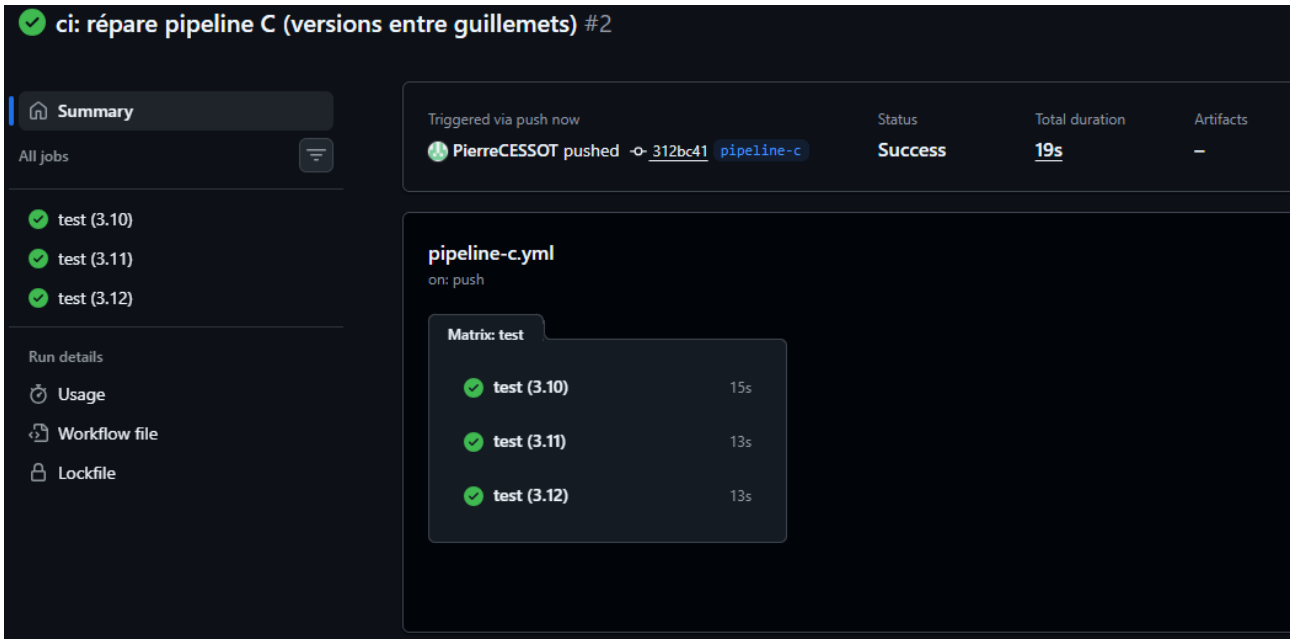

ci: ajoute pipeline A (cassé) #1
Summary
All jobs
Run details
Usage
Workflow file
Lockfile
Workflow file for this run
.github/workflows/pipeline-a.yml at bce22ba
1 name: Pipeline A
2 on: push
3
4 jobs:
5 test:
6 runs-on: ubuntu-latest
7 steps:
Check failure on line 7 in .github/workflows/pipeline-a.yml
GitHub Actions / .github/workflows/pipeline-a.yml
Invalid workflow file
You have an error in your yaml syntax on line 7
8 - uses: actions/checkout@v6
9 - uses: actions/setup-python@v7
10 with:


Pipeline A
ci: répare pipeline A (indentation) #2
Summary
All jobs
test
Run details
Usage
Workflow file
Lockfile
Triggered via push now
PierreCESSOT pushed d57b856 pipeline-a
Status
Success
Total duration
13s
Artifacts
–
pipeline-a.yml
on: push
test 10s


ci: ajoute pipeline B (cassé) #1
Re-run
Summary
All jobs
test
Run details
Usage
Workflow file
Lockfile
Annotations
1 error and 1 notice
test
failed 6 minutes ago in 10s
Search logs
Explain error
Set up job
Run actions/setup-python@v7
Run pip install -r requirements-dev.txt
1 ▶ Run pip install -r requirements-dev.txt
11
12 [notice] A new release of pip is available: 26.2.2 -> 26.3
13 [notice] To update, run: pip install --upgrade pip
14 ERROR: Could not open requirements file: [Errno 2] No such file or directory: 'requirements-dev.txt'
15 Error: Process completed with exit code 1.


ci: répare pipeline B (checkout manquant) #2
Summary
All jobs
test
Run details
Usage
Workflow file
Lockfile
Triggered via push now
PierreCESSOT pushed f1b23b9 pipeline-b
Status
Success
Total duration
15s
Artifacts
–
pipeline-b.yml
on: push
test
12s




Flake8 verifie dans un fichier que le code est utilisé et qu'il n'y a pas d'erreur,
Black réécrit un fichier pour qu'il respecte les style defini,
Si on les sépare ca permet de voir ou ca casse plus vite

Le cache sert a gagner du temps car il garde les donner
un artéfact est un fichier produit par un job

Pour limiter les risques

Git ne peu pas choisir un branche en priorité

Ce sont des données sensible donc il les garde caché
variable = tout le monde peu y acceder
sécret = pas tout le monde

TP3

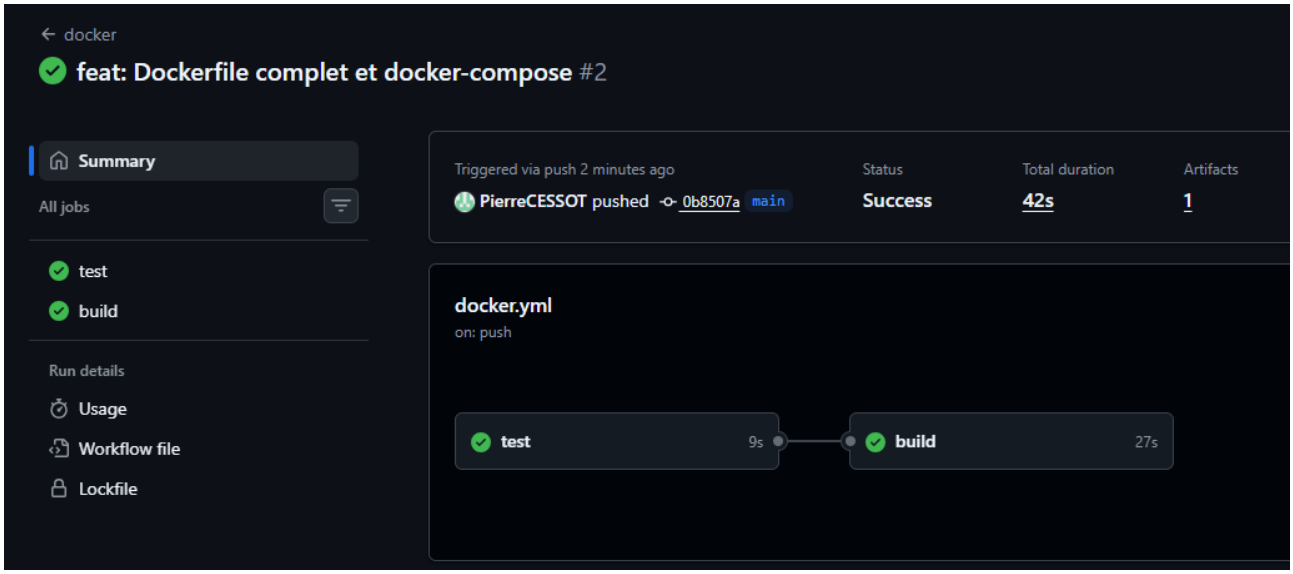
```
REPOSITORY   TAG       IMAGE ID       CREATED          SIZE
tasks-api    local     144d2bb4ea69   3 minutes ago    201MB
```

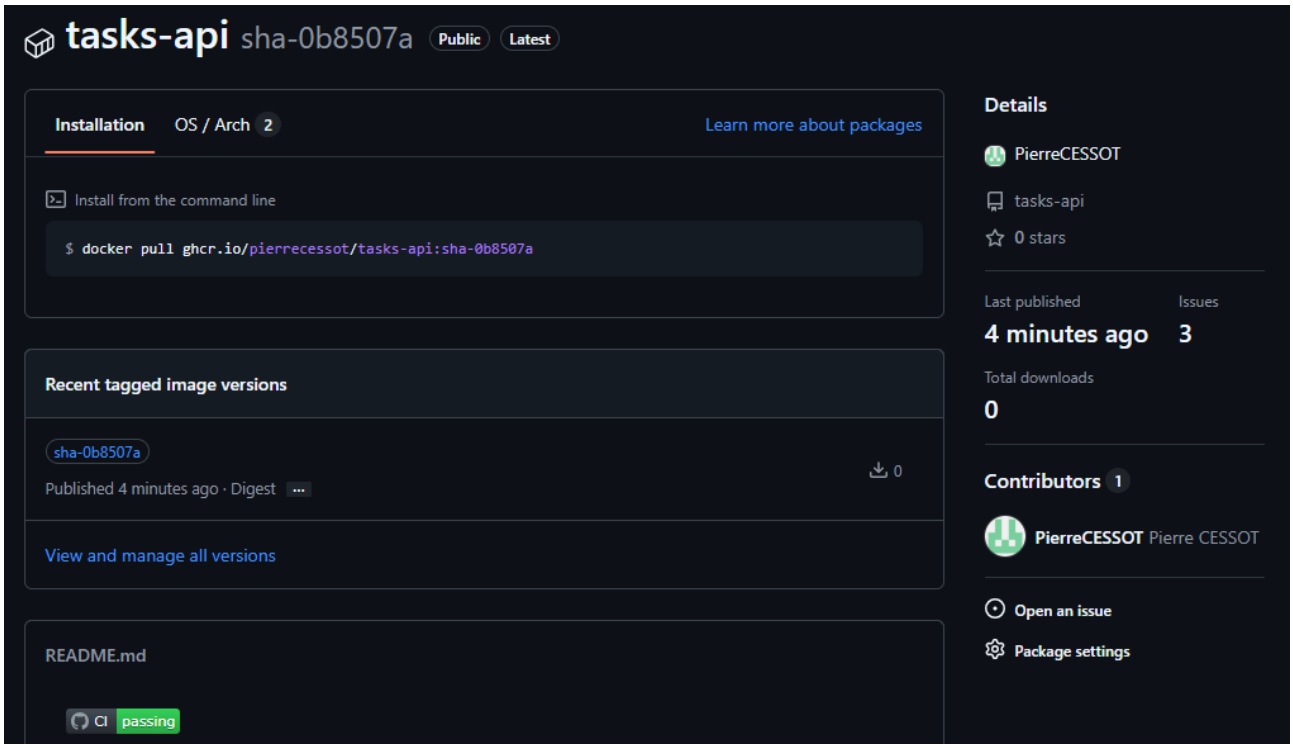
```
PS C:\Users\cesso\Documents\programmation_bas_niveau\exercices\GitOps_CICD\tasks-api> curl.exe http://localhost:8000/health
{"status":"ok"}
PS C:\Users\cesso\Documents\programmation_bas_niveau\exercices\GitOps_CICD\tasks-api> curl.exe http://localhost:8000/version
{"version":"dev"}
PS C:\Users\cesso\Documents\programmation_bas_niveau\exercices\GitOps_CICD\tasks-api>
```

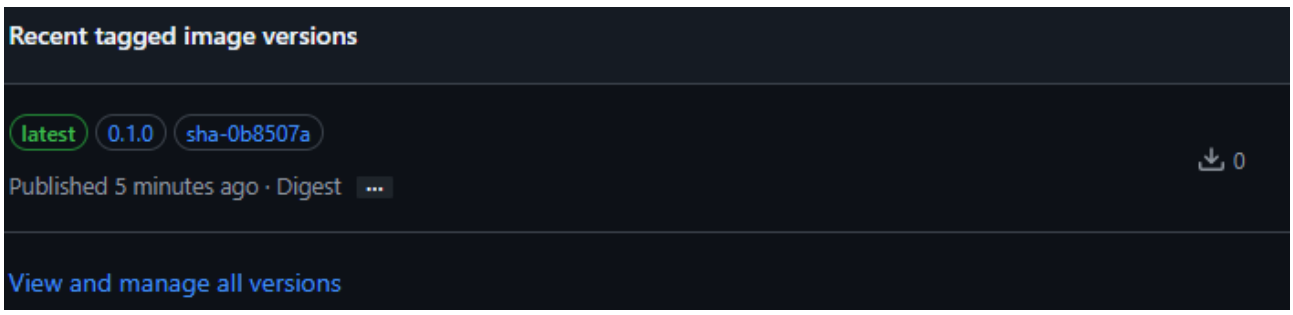
```
 => CACHED [2/6] WORKDIR /app
 => CACHED [3/6] COPY requirements.txt .
 => CACHED [4/6] RUN pip install --no-cache-dir -r requirements.txt
```

```
PS C:\Users\cesso\Documents\programmation_bas_niveau\exercices\GitOps_CICD\tasks-api>    docker compose up
Attaching to api-1esktop   o View Config   w Enable Watch
api-1  | [2026-11-04 12:45:49 +0000] [1] [INFO] Starting gunicorn 26.2.0
api-1  | [2026-11-04 12:45:49 +0000] [1] [INFO] Listening at: http://0.0.0.0:8000 (1)
api-1  | [2026-11-04 12:45:49 +0000] [1] [INFO] Using worker: sync
api-1  | [2026-11-04 12:45:49 +0000] [7] [INFO] Booting worker with pid: 7
api-1  | [2026-11-04 12:45:49 +0000] [1] [INFO] Control socket listening at /home/appuser/.gunicorn/gunicorn.ctl
```

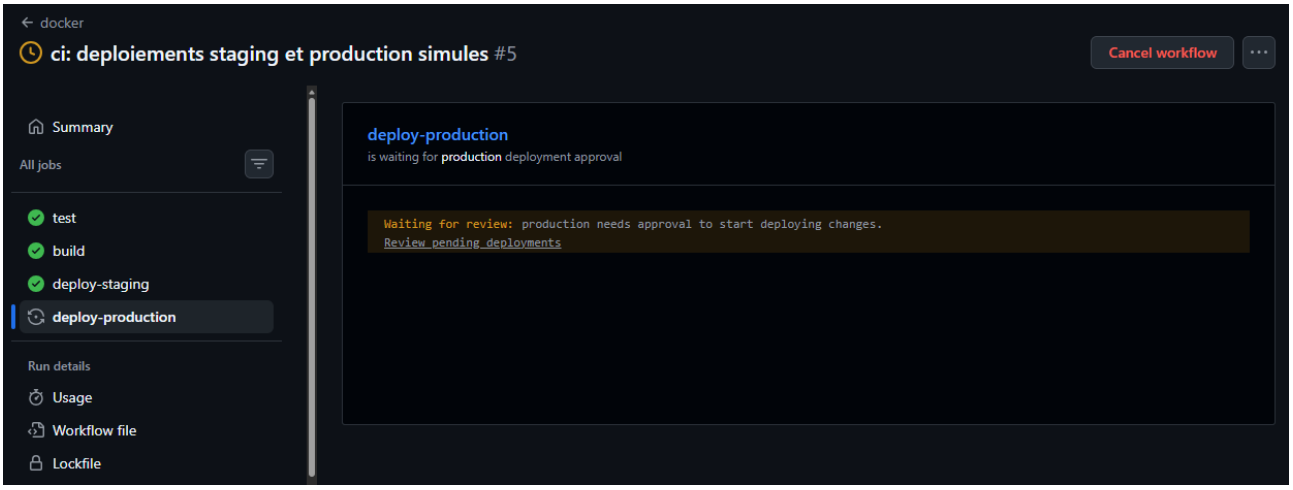
```
PS C:\Users\cesso>    curl.exe http://localhost:8000/version
{"version":"local"}
PS C:\Users\cesso>
```

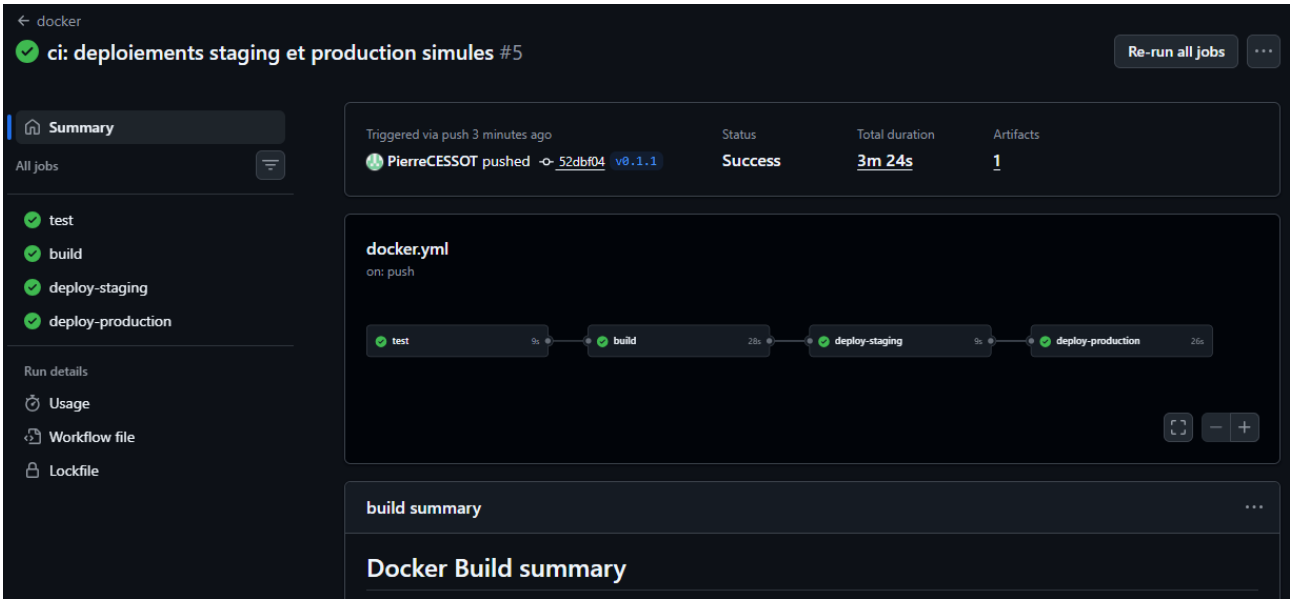





```
PS C:\Users\cesso> curl.exe http://localhost:8000/version
{"version":"0.1.0"}
PS C:\Users\cesso>
```


docker
ci: deploiements staging et production simules #5
Cancel workflow
Summary
All jobs
test
build
deploy-staging
deploy-production
Run details
Usage
Workflow file
Lockfile
deploy-production
is waiting for production deployment approval
Waiting for review: production needs approval to start deploying changes.
Review pending deployments




```
PS C:\Users\cesso\Documents\programmation_bas_niveau\exercices\GitOps_CICD\tasks-api> docker run --rm tasks-api:local id
uid=10001(appuser) gid=10001(appuser) groups=10001(appuser)
PS C:\Users\cesso\Documents\programmation_bas_niveau\exercices\GitOps_CICD\tasks-api>
```

Une image contient les données qui sont utilisé par le conteneur

Le fichier docker-compose.yml permet de lancer la même commande avec les même paramètre

Il faut eviter le tag « latest » en production car les dernieres image peuvent contenir des probleme

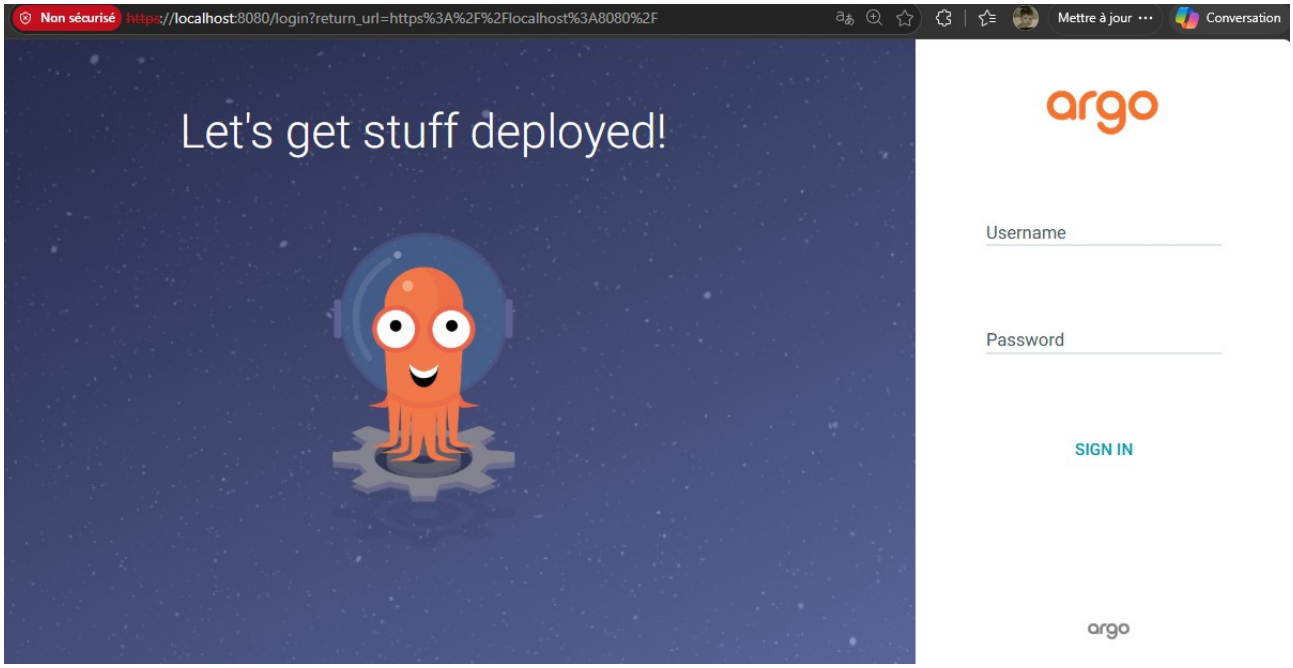
Le test de fumée verifie que l’application demarre et que tout est ok

Un conteneur n’est pas une machine virtuelle

TP4

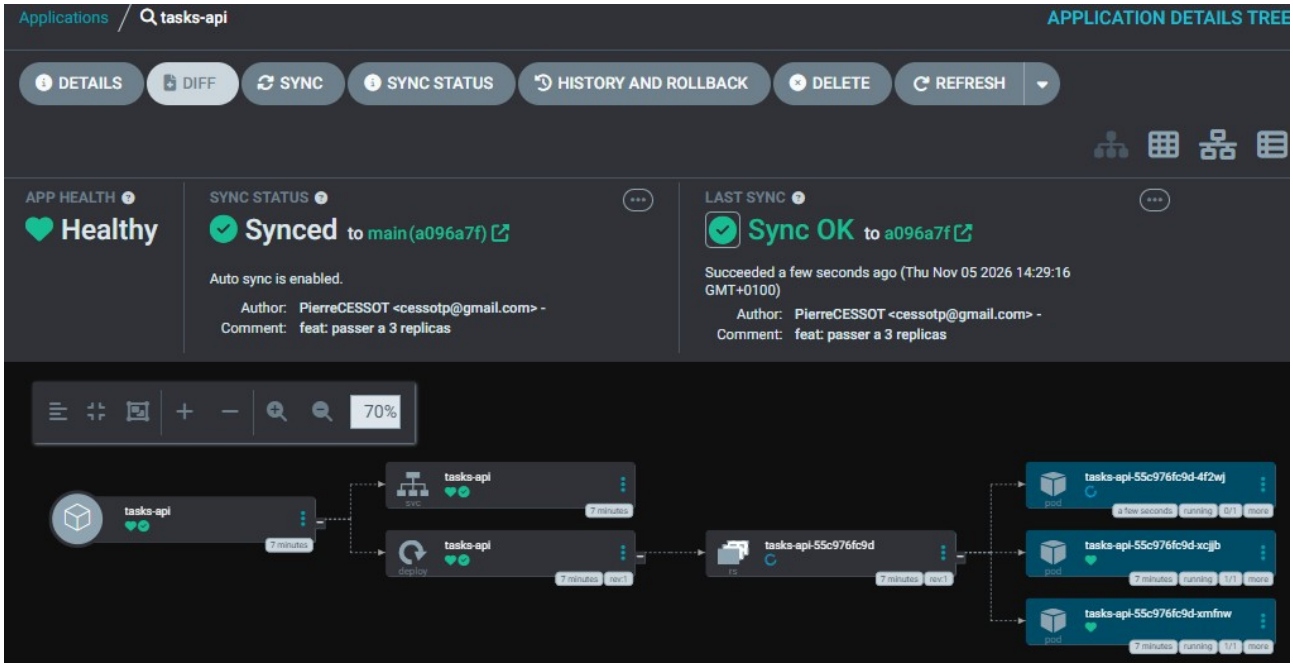
```
PS C:\Users\cesso\Documents\programmation_bas_niveau\exercices\GitOps_CICD\tasks-api> kubectl get nodes
NAME                 STATUS   ROLES           AGE     VERSION
gitops-control-plane   Ready    control-plane   3h12m   v1.37.0
PS C:\Users\cesso\Documents\programmation_bas_niveau\exercices\GitOps_CICD\tasks-api>
```

```
PS C:\Users\cesso\Documents\programmation_bas_niveau\exercices\GitOps_CICD\tasks-api> kubectl get pods -n argocd
NAME                                                READY   STATUS    RESTARTS   AGE
argocd-application-controller-0                     1/1     Running   0          3m54s
argocd-applicationset-controller-794bb95c9d-zw625   1/1     Running   0          3m54s
argocd-dex-server-66cf64bfcf-lxhzh                  1/1     Running   0          3m54s
argocd-notifications-controller-5878c8d644-zzl5b    1/1     Running   0          3m54s
argocd-redis-5869b7547-lc5j4                        1/1     Running   0          3m54s
argocd-repo-server-7d55cc466b-km492                 1/1     Running   0          3m54s
argocd-server-85b849c688-kkt4q                      1/1     Running   0          3m54s
PS C:\Users\cesso\Documents\programmation_bas_niveau\exercices\GitOps_CICD\tasks-api>
```

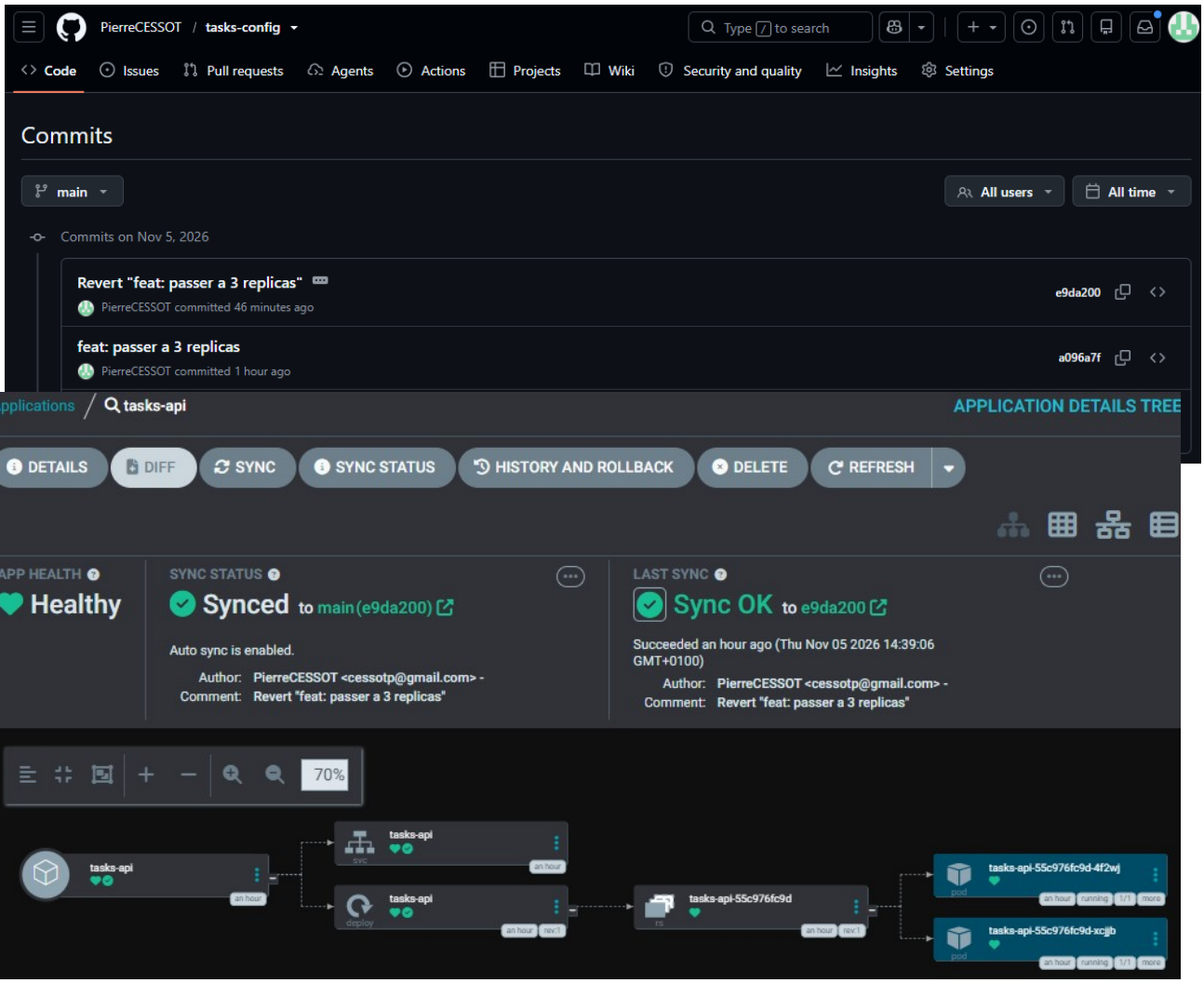





```
PS C:\Users\cesso> curl.exe http://localhost:8082/version
{"version":"0.1.0"}
PS C:\Users\cesso>
```

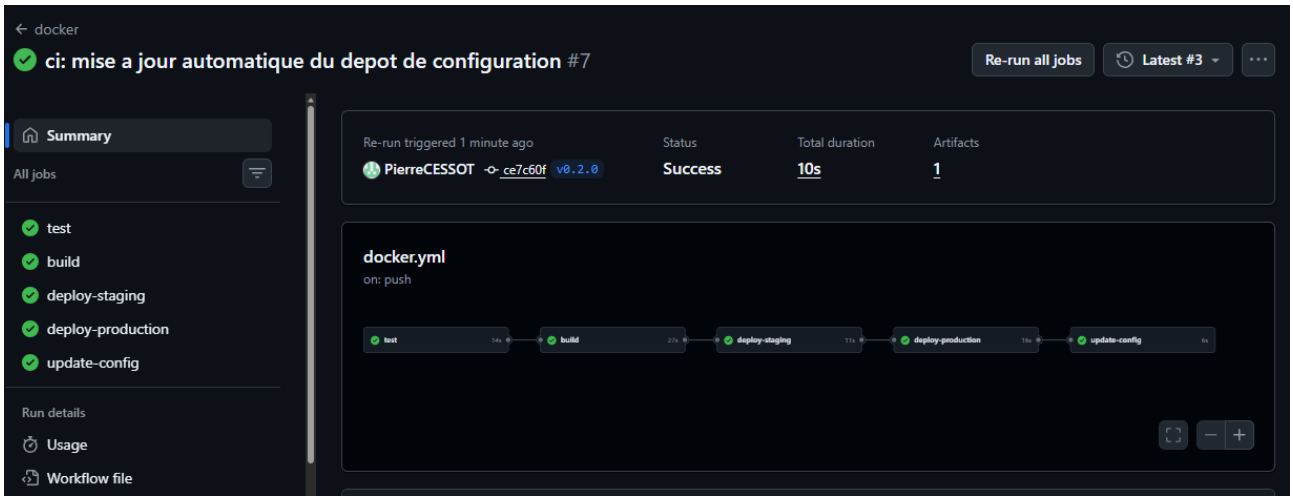


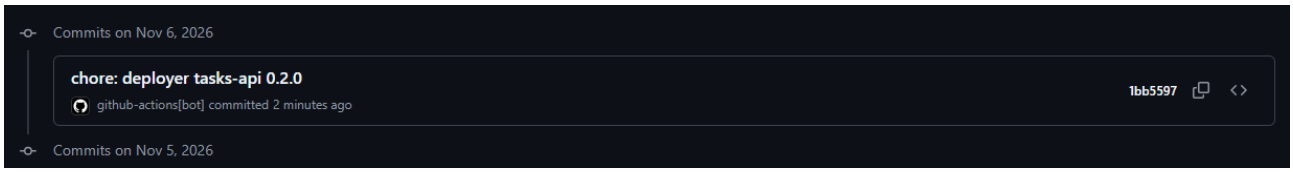

```
PS C:\Users\cesso\Documents\programmation_bas_niveau\exercices\GitOps_CICD\tasks-config> kubectl get pods -n tasks -w
NAME                         READY   STATUS    RESTARTS   AGE
tasks-api-55c976fc9d-4f2wj   1/1     Running   0          5m29s
tasks-api-55c976fc9d-xcjjb   1/1     Running   0          11m
tasks-api-55c976fc9d-xmfnw   1/1     Running   0          11m
tasks-api-55c976fc9d-d7mrc   0/1     Pending   0          0s
tasks-api-55c976fc9d-d7mrc   0/1     Pending   0          0s
tasks-api-55c976fc9d-4nhkp   0/1     Pending   0          0s
tasks-api-55c976fc9d-4nhkp   0/1     Pending   0          0s
tasks-api-55c976fc9d-d7mrc   0/1     ContainerCreating   0          0s
tasks-api-55c976fc9d-4nhkp   0/1     ContainerCreating   0          0s
tasks-api-55c976fc9d-d7mrc   0/1     Terminating         0          1s
tasks-api-55c976fc9d-4nhkp   0/1     Terminating         0          1s
tasks-api-55c976fc9d-4nhkp   0/1     Terminating         0          1s
tasks-api-55c976fc9d-d7mrc   0/1     Terminating         0          1s
tasks-api-55c976fc9d-4nhkp   0/1     Terminating         0          1s
tasks-api-55c976fc9d-d7mrc   0/1     Terminating         0          1s
tasks-api-55c976fc9d-4nhkp   1/1     Terminating         0          6s
tasks-api-55c976fc9d-d7mrc   1/1     Terminating         0          6s
```


PierreCESSOT / tasks-config
Type / to search
Code
Issues
Pull requests
Agents
Actions
Projects
Wiki
Security and quality
Insights
Settings
Commits
main
All users
All time
Commits on Nov 5, 2026
Revert "feat: passer a 3 replicas"
PierreCESSOT committed 46 minutes ago
e9da200
feat: passer a 3 replicas
PierreCESSOT committed 1 hour ago
a096a7f


Applications
tasks-api
APPLICATION DETAILS TREE
DETAILS
DIFF
SYNC
SYNC STATUS
HISTORY AND ROLLBACK
DELETE
REFRESH
APP HEALTH
Healthy
SYNC STATUS
Synced to main (e9da200)
Auto sync is enabled.
Author: PierreCESSOT <cessotp@gmail.com> -
Comment: Revert "feat: passer a 3 replicas"
LAST SYNC
Sync OK to e9da200
Succeeded an hour ago (Thu Nov 05 2026 14:39:06 GMT+0100)
Author: PierreCESSOT <cessotp@gmail.com> -
Comment: Revert "feat: passer a 3 replicas"
70%
tasks-api
tasks-api
tasks-api
tasks-api-55c976fc9d
tasks-api-55c976fc9d-4f2wj
tasks-api-55c976fc9d-xcjjb

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






```
PS C:\Users\cesso\Documents\programmation_bas_niveau\exercices\GitOps_CICD\tasks-api> curl.exe http://localhost:8082/version
{"version":"0.2.0"}
```

Un pod regroupe 1 ou plusieurs conteneurs qui tournent ensemble.

Le depot de l’application declanche la CI a chaque push

Au depart 3 pods tournaient mais apres le kubectl scale à 5, 2 nouveaux pods sont apparus, puis ont été supprimer par la suite

selfHeal = si quelqu’un modifie le cluster à la main, Argo CD le remet automatiquement comme dans Git

Pour revenir en arriere il suffit de faire un git revert du dernier commit